Continuity checklist — stage 1 (unpowered)

Power off. No HV. Use a DMM continuity / resistance mode.
Expected nets: `probable-netlist.md` · record results in `pin-reconciliation.md` · schematic: `../hardware/pmt-interface/pmt-interface.kicad_sch`.

Prep

- ☐ Open datasheet pin circle: `datasheets/XP3312.pdf` (also Prutchi Fig. 4 in `datasheets/Prutchi-XP3312-PMT-Probe.pdf`)
- ☐ Open backlight photos: IMG_7020 (header side), IMG_7005 / 7014 (component side)
- ☐ Mark physical **pin 1** on J1 relative to coax J2
- ☐ **Count J1 pins** (probable **14 = 2×7**; write actual: ____)

1. Tube orientation (datasheet)

Locate **sp**, then confirm:

Tube pin	Expect	OK?
1	D1	
3	D2	
4	D3	
7	D4	
8	D5	
9	D6	
10	D7	
11	D8	
12	A	
16	K	
2,5,6,13–15,17–19	ic / open	

2. Tube → header (probable breakout)

Electrode	Tube pin	Expect J1	OK?
D8	11	**2**	
D7	10	**3**	
D5	8	**4**	
D4	7	**5**	
D3	4	**6**	
K	16	**7** via ≈1 kΩ	
D1	1	**8**	
D2	3	**9**	
D6	9	**12**	
GND	—	**14**	
NC	—	**1, 10, 11, 13**	

3. Coax and anode load

- ☐ **A (12) ↔ coax center** (~0 Ω)
- ☐ Coax shell ↔ board GND ↔ **J1.14**
- ☐ **A → GND ≈ 1.1 MΩ** (R2 115)

4. Cathode passives

- ☐ **K → R1 (1001 / 1 kΩ) → J1.7**
- ☐ **K → C1 (472Z / 2 kV) → GND** (open/charging at DC)

5. Last-dynode bypass cluster (probable)

Expect: `D7 -- C2 --+-- R3 10Ω -- GND` and `D8 -- C3 --+` (shared mid).

- ☐ One tan MLCC between **D7** and mid-node
- ☐ Other tan MLCC between **D8** and same mid-node
- ☐ Mid-node → **100 / 10 Ω** → GND / J1.14
- ☐ If topology differs, sketch actual and update schematic

6. Close out

- ☐ Fill `pin-reconciliation.md`
- ☐ Edit KiCad nets to match reality; strip wrong probabilities
- ☐ Re-export PDF if desired: `kicad-cli sch export pdf hardware/pmt-interface/pmt-interface.kicad_sch -o docs/pmt-interface-draft.pdf`